

Amol Dadarao Rathod

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TECHNICAL SKILLS

- **Languages:** Python, C++, C
- **Core Concepts:** Data Structures & Algorithms, OOPs, Operating Systems, Computer Networks, System Design
- **Domains:** Artificial Intelligence (LLMs, RAG, NLP), Data Science, Machine Learning, Deep Learning, Cloud Computing, Web Development
- **Tools & Databases:** Vector Databases, Oracle Autonomous Database, Git/GitHub, n8n, Oracle Cloud, Docker, Hugging Face, Vercel

EXPERIENCE

VR Coatings Pvt. Ltd. — AI Intern

Pune, Maharashtra — Sep 2026 – Present

- Developing a **RAG-based AI system** for VR Coatings to enable intelligent retrieval and question answering over company-specific information.

PROJECTS

Vidhoor - Legal AI Copilot — [Live Demo](#) — [GitHub](#)

Mar 2026 – Present

Tech Stack: Llama 3.1 8B (Cerebras), RAG, Vector DB, Oracle DB

- Engineered an AI-powered legal assistant utilizing **Retrieval-Augmented Generation (RAG)** and Vector Databases, delivering real-time insights with ultra-low latency inference speeds of **2,350 tokens per second**.
- Processed, trained, and indexed **851 pages** of comprehensive Indian legal frameworks, including the Indian Constitution, BNS, BNSS, IPC, IT Law 2000, and Data Privacy Laws.
- Hosted the architecture on **Oracle Autonomous Database** leveraging Hybrid Search capabilities.
- Implemented strict data security measures, including End-to-End Encryption (E2EE) and Personally Identifiable Information (PII) handling/redaction.

Interview.io - AI Interview Prep Platform — [GitHub](#)

Jan 2026 – Present

Tech Stack: Llama 3.1 8B (Cerebras), Express.js, React.js

- Developed a career preparation platform enabling students to practice via **AI-driven mock interviews** tailored to specific technical domains like Software Engineering and Data Science.
- Built an automated resume parser that provides instant, actionable feedback on formatting and keyword optimization.
- Integrated a “Codex” module featuring curated Data Structures and Algorithms (DSA) questions covering arrays, trees, and dynamic programming.
- Designed a customized roadmap generation feature that curates tailored study resources and reference links based on the user’s career goals.

Sovereign AI - Air Gapped Workbench — [GitHub](#)

2026 – Present

Tech Stack: LangGraph, RAG, Qdrant, BM25, FastAPI, PostgreSQL, Docker

- Engineered a modular **agentic AI system** using LangGraph with specialized workflows for planning, retrieval, analysis, calculation, decision-making, generation, synthesis, and verification.
- Implemented a **hybrid RAG pipeline** combining dense vector retrieval with BM25 sparse retrieval and weighted fusion to improve contextual document retrieval.
- Built an autonomous coding agent with a **generate–test–analyze–fix** workflow and sandboxed code execution with restricted network, filesystem, subprocess, and environment access.
- Containerized the AI platform using **Docker** and integrated FastAPI, PostgreSQL, Qdrant, and local LLM inference into a modular, production-oriented architecture.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** [Career Essentials in Generative AI \(Microsoft\)](#), [GitHub Foundations Part 1](#) & [Part 2](#), [Supervised & Unsupervised Machine Learning](#).
- **Finalist**, OpenAI Academy x NxtWave Buildathon with **AI Receptionist**.
- **Finalist in 5+ hackathons**, demonstrating experience in building and presenting AI-driven solutions under competitive timelines.
- Active competitive programmer with 200+ problems solved on LeetCode focusing on Data Structures and Algorithms.

EDUCATION

P.E.S. Modern College of Engineering

Pune, MH

B.Tech. in Artificial Intelligence and Data Science

July 2024 – Expected July 2028

Current Year: Third Year (T.Y B.Tech)

CGPA: 9.00

Key Coursework: Data Structures (DSA), Object-Oriented Programming (OOP), Operating System (OS).